Business Case
For PukParking

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	Information
Document ID	*[Document Management System #]*
Document Owner	*[Owner Name]*
Issue Date	*[Date]*
Last Saved Date	*[Date]*
File Name	*[Name]*

Document History

Version	Issue Date	Changes
[1.0]	*[Date]*	*[Section, Page(s) and Text Revised]*

Document Approvals

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1 Executive Summary

Currently the North-West University (NWU) Potchefstroom campus faces a significant parking challenge, with students, guests and even staff often required to drive around campus searching for an available parking spot, just to end up parking in a spot that does not correspond with their disc type due to faded markings, which could result in difficult to pay fines.
PupParking addresses this issue through a mobile-first Progressive Web Application (PWA) designed to help its users find the closest available and valid parking spot to their desired destination on campus. The recommendation algorithm considers things such as walking distance, parking disc type, and how many users reported a parking as occupied.

When users report a parking area as full, the system temporarily reduces its recommendation priority and recalculates an alternative parking area, helping distribute users across available parking without requiring dedicated parking sensors or user accounts.

Three solutions were considered:

(1) PukParking, a custom-developed campus parking application
(2) An existing parking application, which could reduce development effort but introduce licensing, customisation and vendor-dependency limitations
(3) a Smart parking sensor system, which would provide automated real-time occupancy information but require significant infrastructure and maintenance.

PukParking was found to be the most feasible solution, receiving an overall feasibility rating of 8.8/10, compared with 7.4/10 for an existing parking application and 7.2/10 for the smart parking sensor system. It also achieved the highest solution rating of 54/60. PukParking is therefore recommended because it can be tailored to NWU's requirements, provides greater system control, reduces vendor dependency and allows future expansion.

Implementation will follow a hybrid Design Science Research Methodology (DSRM) and Agile approach, using two-week sprints for incremental development, testing and stakeholder feedback. Development will progress from the system foundation and campus data through mapping, parking recommendations, navigation, crowd-assisted re-routing and PWA functionality, followed by testing and deployment.

The implementation will focus on the core parking-discovery and navigation features while avoiding the initial cost and complexity of physical sensors. Following deployment, administrator training, documentation, knowledge transfer and user acceptance testing will support operational use, while project progress, quality, costs, risks and issues will be monitored throughout implementation.

2 Business Problem

2.1 Environmental Analysis

The North-West University (NWU), Potchefstroom Campus, is one of South Africa's largest contact university campuses. The NWU's own institutional statistics report, reports that in 2021 it was the fourth-largest university in South Africa, according to the total headcount, with 53,469 students enrolled for 2021 (North-West University, 2024). Given this scale, it has long created land and traffic pressure. A formal transport study of the campus and its surroundings have found that volume-to-capacity ratios at several inter- and intra-campus survey points exceeded 0.95, which the researchers identify as a clear indicator of traffic congestion on campus roads (Schoeman, 2014). The same research programme's parking survey, covering 63 survey points across the main campus in Potchefstroom, recorded a total of 4,342 parking spaces available, of which 923 were undercover parking, reserved for staff and 2,487 were open parking for staff and students (Schoeman, 2014). While this data could currently be inaccurate due to the survey being done 12 years ago, it establishes that campus parking demand at the NWU Potchefstroom Campus has for many years been very high in many allocated parking zones. The environmental context is unpacked below in terms of the macro-level (PESTEL) and micro-level environmental analysis forces that influence its need for the PukParking project.

Macro-level Environmental Analysis (PESTEL)

1. **Political**

Given the NWU's Digital version of 2023's Business Strategy, approved by Council in 2023, naming digital transformation as one of six strategic institutional goals, also creating institutional backing for digitising manual, campus-facing processes, in this case including the parking on campus (North-West University, 2023).

Opportunity: An approved institutional strategy was already legitimatising digital projects such as this one.

Threat: the budget or governance priorities could shift focus towards other digital innovations; therefore, this innovation can possibly not approve.

2. **Economic**

The PukParking project is designed to run at effectively zero cost. The frontend/backend stack (React, Node.js, Express, MongoDB) are open-source

software, and the mapping layer which will be used relies on Mapbox, whose free tier offers roughly 50,000 web map loads per month, plus separate free monthly allowances for geocoding and directions requests in the tens of thousands, comfortably covers a single-campus deployment with no paying customer base (Radar, 2026).

Opportunity: A fully functional Minimum Viable Product (MVP) can be built and hosted within an academic project's zero/near-zero budget.

Threat: If usage grows beyond the free tier as offered by Mapbox, Mapbox's usage-based pricing (per-1,000 map loads, requests, or monthly active users) would introduce a recurring cost NWU, which is not budgeted for.

3. **Social**

Given a report on South Africa’s digital footprint, South Africa had 50.8 million internet users at the start of 2025, an internet penetration rate of 78.9% of the population, with continued year-on-year growth in mobile internet use (DataReportal, 2025). This supports the assumption that most students, staff and visitors will be able to access this browser-based Progressive Web App (PWA) without needing to download a native app on their device.
Opportunity: High digital readiness of users lowers adoption struggles for this no-login PWA.

Threat: A minority of prospective users, particularly cost-sensitive students and occasional visitors to the campus, may still have limited data or older devices not compatible with this PWA, therefore the application must remain lightweight and cache well for low-bandwidth conditions.

4. **Technology**

The NWU already offers some self-service digital tools such as the DIY Services portal's parking-disc and registered-vehicle apps, showing institutional and student familiarity with app-based campus services, but unfortunately these only display a user's own existing disc/vehicle details rather than offering navigation to various available parking. Established, well-documented mapping technology such as Mapbox GL JS, Directions API, Geocoding API makes it possible for developers to build interactive campus navigation without any in-house mapping infrastructure that needs to be set up for the technology to work.

Opportunity: The required mapping and PWA technologies are mature, free at this scale, and require no new campus hardware that must be adopted.

Threat: The PukParking application depends on a third-party mapping provider and a stable connectivity for the best usage and experience. If a period of network disruption could occur, it could degrade the live-navigation features, which the PWA's offline caching is intended to partially solve if such problem could occur.

5. **Environmental**

The NWU has an approved Environmental Sustainability Policy aligned with national green initiatives (North-West University, 2022). The reduction of the amount of time in vehicles spent circling campus roads to find parking, that are already shown to run close to capacity (Schoeman, 2017) would reduce lower vehicle emissions and unnecessary fuel waste.

Opportunity: A parking navigation tool such as PukParking that shortens search time for parking on campus directly supports the NWU's sustainability goals as discussed.

6. **Legal**

The Protection of Personal Information Act 4 of 2013 (POPIA) governs any personal information the NWU processes of students, staff, visitors, and the NWU has an active Personal Information Privacy Policy and Information Governance Framework overseeing compliance of this act (North-West University, 2026d). Because PukParking's end-user features (map browsing, search, navigation, "Parking Full" reporting) are designed to work without registration or login from users, the public facing application collects no personal information from students, staff, or visitors, significantly limiting its POPIA exposure.

Opportunity: A no-login design supports the fact that user’s personal information will be stored, therefore limiting its POPIA exposure.

Threat: The administrator portal of PukParking does process staff login credentials, thus standard safeguards for applications such as password hashing, JWT-based authentication and role-based access remain necessary when developing the application. Any future feature that introduced user accounts or individual location history would substantially increase POPIA compliance obligations.

Micro-level Environmental Analysis

1. **Customers**

The primary customers of PukParking will be the NWU’s students, staff, and visitors, who currently have no way to find a valid, available parking area or spot near their destination of choice. Students and staff who were interviewed about the current disc-based system, described inconsistent success in finding parking and confusion

among newer students about which zones their disc permits them to park was also mentioned (Coetzee & Du Plessis, 2022). Secondary customers are the NWU's Traffic and Access Control department and campus administrators, who need a simple way to manage building and parking-area data.

2. **Competitors / Existing Solutions**

General-purpose navigation apps such as Google Maps, Waze, Apple Maps etc. can route a user to a named NWU parking area as shown by public Waze or Google Maps listings for specific NWU parking zones, but these general-purpose navigation apps have no knowledge of disc-zone restrictions, current occupancy, or which area is closest to a specific building, they only navigate to a fixed location rather than recommending the best valid parking spot. The NWU's own DIY Services portal only displays a user's already-issued parking disc and vehicle details as previously discussed and its official parking guidance only points users to a non-interactive PDF version of the campus maps. None of these existing options combine destination-based recommendation, zone-restriction awareness, and real-time parking availability.

3. **Suppliers**

Mapbox is the key technology supplier for mapping, geocoding, and route generation in this project, operating on a usage-based pricing model with a free tier suited to a single-campus academic deployment. The remainder of the stack, software Node.js, Express, MongoDB, React and related open-source libraries, has no licensing supplier dependency at all, which keeps the project's technology costs close to zero.

4. **Substitutes**

The current substitute as provided by the NWU, is the NWU's existing manual process of printed purchased parking discs, colour-coded signage, and word-of-mouth or Traffic and Access Control staff guidance (Coetzee & Du Plessis, 2022). General consumer navigation apps are a partial substitute for reaching a named parking area, but not for finding an available, permitted parking spot. Therefore, none of these substitutes offer live occupancy awareness or destination-based recommendation for the NWU’s students, staff and visitors.

5. **New Technology**

A sensor-based "smart parking" system such as using ground sensors or cameras to detect real-time occupancy of parking spots, represents a more capital-intensive alternative that the NWU has not committed hardware budget toward. Alternatively,

PukParking's crowd-sourced "Parking Full" reporting model offers a comparable live-availability signal without requiring new campus hardware, positioning it as a low-cost entrant relative to sensor-based alternatives.

2.2 Problem Analysis

Business Problem

The main problem that is applicable in this project is, that the NWU Potchefstroom Campus has no real-time, guided way for students, staff, or visitors to find the nearest valid and available parking area for a specific destination they want to go to.

The cause of this problem is:

- Structural parking pressure: The historical campus parking surveys (as previously mentioned) show utilisation of parking areas regularly reaching 82–92% in key zones, meaning users are often searching in areas that are either already full or close to full (Schoeman, 2017).
- Manual sign-based permission system: Where students, staff and visitors may park depends on their disc colour and zone, communicated only through the physical windscreen discs and signage boards around campus, with no digital lookup of what is currently permitted or available to the students, staff and visitors (Coetzee & Du Plessis, 2022).
- No real-time parking guidance: The NWU's own resources point users to a PDF version of the campus map rather than a tool that accounts for current occupancy or restrictions applicable on the campus (North-West University, 2026a).
- No accommodation for visitors: First-time or occasional campus visitors have no way to guide themselves or be guided to their destinations without any prior knowledge of the zone system.

The impact of this problem include, unnecessary vehicle movement on campus while students, staff and visitors search for parking, a risk of receiving a parking fine is also applicable for students, staff and visitors who misjudge zone restrictions, and a poor first impression for visitors navigating the campus for the first time.

The aim to have the given business problem be solved and completed, is by 16 October 2026.

Business Opportunity

Digitalising the parking allocation process will present an opportunity to the NWU in the following way:

- Reducing the time students, staff and visitors spend circling the campus searching for available parking by using destination-based recommendations, instead of relying on current signage.
- Reducing unnecessary traffic movement on campus and congestion on and off campus in the areas already shown to run close to capacity (Schoeman, 2017).
- Give visitors and first-time users an easily accessible way to navigate campus without needing to learn the disc/zone system in advance, since the application requires no prior information to be learnt.
- Align with the NWU's active Digital Business Strategy (as previously mentioned), which prioritises digital transformation as an institutional goal (North-West University, 2023), and with the NWU's existing move toward app-based services such as its parking-disc and registered-vehicle DIY Services apps.
- Provide administrators with a portal to manage real-time parking data, something the current process does not support at all.

The NWU already offers digital, self-service apps for parking-disc and vehicle information, showing that the university and its students and staff are familiar with app-based tools.

This business opportunity would directly reduce search-related congestion and improve the campus experience for the students, staff, and visitors.

3 Alternative Solutions

3.1 Option 1

3.1.1 Description

PukParking is a dedicated mobile and/or web-based campus parking application designed around the needs of North-West University. It would provide students, staff and visitors with a single interface for locating suitable parking and navigating to it. The proposed core functions are:

- View available or designated parking areas.
- Search for parking near a specific campus building or area.
- View parking areas on a campus map.
- Receive directions to selected parking areas.
- Identify parking zones allocated to specific user groups.
- View parking availability where reliable sensor data or controlled manual updates are available.
- Potentially reserve parking spaces where institutional policy permits.

- Receive notifications concerning parking availability, restrictions or changes.

The underlying database would contain campus parking areas, parking spaces, buildings, users and parking rules. This structure allows the application to be configured to NWU's actual parking layout and operating rules rather than requiring the university to adapt its processes to a generic parking product. NWU's published parking information confirms that parking is organised around designated areas, campus maps and user/permit restrictions (North-West University, 2026).

The solution can also be implemented incrementally. A first release can focus on parking discovery, campus mapping and directions, while more infrastructure-intensive features such as automated occupancy detection can be introduced later. This staged approach reduces the need for a campus-wide hardware investment at the initial implementation stage.

3.1.2 Benefits

Benefit Category	Benefit Description	Benefit Value
Campus-specific navigation	Search for parking near a specific building and view designated areas on an NWU-focused map.	Directly addresses the parking-search requirement.
Flexible information model	Parking zones, buildings, rules and user categories can be configured for NWU.	Supports campus-specific rules without depending on a generic product.
Phased implementation	Core parking discovery can be implemented before optional real-time sensor integration.	Reduces the initial hardware and infrastructure requirement.
Future integration	The platform can later accept trusted availability or sensor data.	Provides a practical path to enhanced smart parking.

3.1.3 Costs

Expense Category	Expense Description	Expense Value	Expense Type
Domain name	Basic .co.za or similar domain for a pilot/project website.	*R150–R350 per year*	*Annual*
Hosting and database	Student-friendly Firebase/free-tier hosting and database during the pilot. Firebase provides no-cost quotas on its Spark plan.	*R0–R300 per month*	*Recurring*
Maps and directions	Small pilot usage of mapping/directions services. Google Maps Platform provides free monthly usage thresholds; additional usage is billed by SKU.	*R0–R500 per month*	*Recurring*

Data preparation and testing	Parking data capture, verification, test devices and minor printing/signage.	*R300–R800*	*Once-off*
Development tools	Free/open-source tools such as VS Code, Git and free-tier services.	*R0*	*Once-off*
Estimated student-project setup	Initial project estimate excluding student labour and existing university equipment.	*Approximately R450–R1,650*	*Estimate*

3.1.4 Feasibility

Solution	Feasibility Rating	Assessment Method
PukParking	9/10	Technical feasibility: ability to implement the application, database, mapping and notifications using established technologies.
PukParking	7/10	Economic feasibility: considers development, hosting and maintenance costs; phased delivery reduces initial cost exposure.
PukParking	9/10	Operational feasibility: ease of use, administration and alignment with campus parking processes.
PukParking	9/10	Scalability: ability to add parking areas, campuses, users and future integrations.
PukParking	10/10	User requirements: supports building-based parking search, campus mapping, designated zones and navigation.

Overall feasibility rating: 8.8/10, based on the average of the five criteria.

3.1.5 Risks

Risk Description	Risk Likelihood	Risk Impact	Actions Required to Mitigate Risk

Development and maintenance cost	Medium	High	Develop the solution in phases and prioritise the parking-discovery, mapping and building-search functions first.
Real-time availability	High	Medium	Begin with verified manual updates or existing trusted data sources; introduce sensor integration as a later phase.
Technical failure	Medium	High	Use reliable hosting, backups, monitoring, testing and recovery procedures.
User adoption	Medium	Medium	Provide a simple interface, clear onboarding and user guidance; use feedback to refine the application.

3.1.6 Issues

Issue Description	Issue Priority	Actions Required to Resolve Issue
Parking data must remain accurate and reflect designated areas, restrictions and changes.	High	Establish a controlled process and responsible owner for updating parking information.
The application requires an accurate and maintainable campus map.	High	Obtain approved campus maps and define a process for keeping map data current.
Future integration with access-control, sensor or other parking systems may be required.	Medium	Use modular architecture and documented APIs so future integrations can be added without redesigning the entire application.

3.1.7 Assumptions

- North-West University can provide or approve accurate information about existing parking areas and rules.
- Students, staff and visitors have access to smartphones or web-enabled devices.
- The university can provide or approve the required software infrastructure and hosting arrangements.
- The initial implementation focuses on parking discovery and navigation rather than automated physical parking control.
- Real-time parking detection can be introduced at a later stage when reliable data sources or sensors are available.

3.2 Option 2

3.2.1 Description

The second alternative is to use an existing parking-related application or platform rather than developing a dedicated system from the ground up. The University of Pretoria's mobile

application demonstrates that an institutional mobile platform can provide campus maps, navigation-related information and parking-disc management functions (University of Pretoria, 2024). UP also operates designated parking arrangements and access controls through its security and parking services (University of Pretoria, 2026).

For NWU, this approach would require identifying an existing product that can be configured for NWU's campuses, parking areas and rules. Its principal advantage is a potentially shorter implementation period and reduced internal software-development effort. Its limitation is that NWU would depend on the selected provider or platform for the availability of the required features and the degree of customisation.

3.2.2 Benefits

Benefit Category	Benefit Description	Benefit Value
Faster starting point	Existing mapping and mobile functionality can reduce the amount of software that must be built.	Potentially shortens implementation time.
Reduced development effort	The alternative avoids building every application component from scratch.	Suitable where an existing platform meets most requirements.
Established functionality	Existing platforms may already include maps, navigation and user-facing features.	Reduces technical uncertainty for basic functions.
Lower initial custom development	Configuration may be simpler than creating a new application.	Cost advantage depends on licensing and integration requirements.

3.2.3 Costs

Expense Category	Expense Description	Expense Value	Expense Type
Application/platform licence	No public institutional licence price is assumed for reusing another university's application. A vendor or institutional agreement would be required.	R0–R2,500 per month*	Recurring
Configuration and setup	Basic configuration of campus information, maps, parking areas and user access.	R500–R2,000	Once-off
Data and compatibility testing	Testing campus maps, building search, parking zones and data accuracy.	R300–R800	Once-off
Integration allowance	Small allowance for linking available campus information to the selected platform.	R500–R1,500	Once-off
Training and rollout material	Simple user instructions, demonstrations and rollout material.	R200–R500	Once-off

Estimated student-project setup	Excludes any commercial licence because a public price is not available.	Approximately R1,500–R4,800*	Estimate

3.2.4 Feasibility

Solution	Feasibility Rating	Assessment Method
Existing Parking Application	8/10	Technical feasibility: mature mobile, mapping and parking functionality already exists.
Existing Parking Application	8/10	Economic feasibility: potentially lower implementation effort, subject to licensing and configuration costs.
Existing Parking Application	8/10	Operational feasibility: established platform can reduce implementation effort but depends on provider processes.
Existing Parking Application	7/10	Scalability: expansion depends on the selected platform and available customisation.
Existing Parking Application	6/10	User requirements: may not fully support NWU-specific parking zones, building search and rules.

Overall feasibility rating: 7.4/10, based on the average of the five criteria.

3.2.5 Risks

Risk Description	Risk Likelihood	Risk Impact	Actions Required to Mitigate Risk
Limited customisation	High	High	Conduct detailed requirements analysis and product testing before adoption.
Vendor dependency	Medium	High	Review service-level agreements, support arrangements, data access and exit provisions.
Subscription or licensing costs	Medium	Medium	Compare total cost of ownership over several years before implementation.

Data limitations	Medium	High	Perform data-accuracy and coverage testing using representative NWU campus information.

3.2.6 Issues

Issue Description	Issue Priority	Actions Required to Resolve Issue
The selected application may be designed for public or commercial parking rather than NWU campus navigation.	High	Perform a detailed compatibility assessment against NWU requirements.
The application may not support building-based parking search across NWU campuses.	High	Confirm functionality through product demonstrations, testing and vendor documentation.
Parking data may remain controlled by the service provider.	Medium	Define data ownership, access, portability and retention requirements contractually.

3.2.7 Assumptions

- A suitable existing application can be identified and assessed.
- The provider permits institutional use and configuration for NWU.
- The application can support the geographical areas and campus maps required by NWU.
- The platform can satisfy most core parking-discovery requirements without extensive custom development.
- NWU accepts the level of vendor dependency and data governance associated with the selected platform.

3.3 Option 3

3.3.1 Description

The Smart Parking Sensor System is an IoT-based alternative that uses physical sensors to determine whether individual parking spaces are occupied. Sensor readings are transmitted to a central server and can be displayed through a mobile or web interface. An administrator dashboard can be used to monitor occupancy, manage parking areas and identify faulty sensors.

The solution would consist of parking sensors, wireless communication equipment, a backend server, database, mobile/web application and administrator dashboard. This architecture can provide highly current occupancy information and reduce dependence on manual user reports. Research confirms that sensor-enabled parking systems can transmit occupancy information to online interfaces and help users identify available spaces (Dheeven et al., 2024).

The main feasibility constraint is that the solution introduces physical infrastructure across the parking estate. The project review estimates sensor, network, software, hosting and maintenance costs and identifies campus-wide deployment as Medium-Low feasibility. A pilot in high-demand areas is therefore more practical than immediate full-campus installation.

3.3.2 Benefits

Benefit Category	Benefit Description	Benefit Value
Real-time occupancy	Sensors can automatically detect whether a parking space is occupied.	Provides current availability when the sensor network is functioning reliably.
Reduced manual reporting	Occupancy information is collected automatically.	Reduces dependence on manual user reports.
Parking analytics	Historical occupancy data can be stored and analysed.	Can support peak-period identification and parking planning.
Future integration	Sensor data can be connected to a web or mobile interface.	Provides a strong enhancement option after the core platform is established.

3.3.3 Costs

Expense Category	Expense Description	Expense Value	Expense Type
Ultrasonic sensors	**Low-cost prototype sensors can be used for a small proof of concept. A South African supplier lists the HC-SR04 at R27.60 including VAT.**	**R27.60 per sensor**	**Once-off**
IoT controller	**An Arduino Nano 33 IoT is listed locally at R690 including VAT.**	**About R690 per controller**	**Once-off**
Prototype wiring and enclosures	**Breadboards, cables, power supplies and basic protective enclosures.**	**R300–R800**	**Once-off**
Local processing / gateway	**A Raspberry Pi 4B 4GB is listed locally at R2,587.50 including VAT and can support a small prototype.**	**About R2,587.50**	**Once-off**
Cloud/database	**A small prototype can use Firebase's no-cost tier; paid usage applies after quotas are exceeded.**	**R0–R300 per month**	**Recurring**
Estimated 10-space pilot	**Approximate small proof-of-concept using low-cost sensors, controllers and basic materials.**	**Approximately R3,900–R5,500**	**Pilot estimate**

3.3.4 Feasibility

Solution	Feasibility Rating	Assessment Method
Smart Parking Sensor System	8/10	Technical feasibility: sensors, communications, backend and interface are established technologies.
Smart Parking Sensor System	7/10	Economic feasibility: physical sensors, networking, installation and maintenance increase initial cost.
Smart Parking Sensor System	7/10	Operational feasibility: provides useful real-time information but requires infrastructure maintenance.
Smart Parking Sensor System	6/10	Scalability: campus-wide expansion requires additional hardware, connectivity and maintenance.
Smart Parking Sensor System	8/10	User requirements: directly supports automated, real-time parking occupancy information.

Overall feasibility rating: 7.2/10, based on the average of the five criteria.

3.3.5 Risks

Risk Description	Risk Likelihood	Risk Impact	Actions Required to Mitigate Risk
Sensors may malfunction.	Medium	High	Implement sensor monitoring, scheduled maintenance and replacement procedures.
High initial implementation cost.	High	High	Start with a limited pilot in high-demand areas before wider deployment.
Sensors may be damaged or vandalised.	Medium	High	Use durable equipment and protective installation methods.
Wireless connectivity may fail.	Medium	High	Conduct network testing and install additional gateways where required.
Sensor batteries may run out.	Medium	Medium	Schedule installation during low-traffic periods and coordinate with campus parking management.

3.3.6 Issues

Issue Description	Issue Priority	Actions Required to Resolve Issue
Installing sensors in every parking space can be expensive.	High	Begin with high-demand parking areas and expand only after pilot results justify wider deployment.
Sensors require physical maintenance.	Medium	Establish regular inspection, maintenance and replacement schedules.
Existing campus network infrastructure may not support all sensors.	High	Conduct an infrastructure and network assessment before deployment.
Sensor batteries may run out.	Medium	Schedule installation during low-traffic periods and coordinate with campus parking management.

3.3.7 Assumptions

- The university has sufficient funding for the required IoT equipment and supporting infrastructure.
- Parking spaces can be physically modified for sensor installation.
- Suitable wireless connectivity can be provided throughout the pilot and later deployment areas.
- Sensors can operate reliably under campus environmental conditions.
- Staff or contracted personnel will be available to maintain the sensor infrastructure.
- The university approves the installation of IoT equipment.
- A pilot implementation will be completed before full deployment.

4 Recommended Solution

4.1 Solution Rating

Assessment Criteria	Solution 1 Rating(1-5)	Solution 2 Rating(1-5)	Solution 3 Rating(1-5)
Alignment with user requirements	5	5	2
Real-time data accuracy	3	4	5
Audit trailing and reporting	5	5	5
NWU policy compliance	5	3	2
Scalability	4	3	5
Maintenance	5	5	3
Usability/User experience	5	5	3

Platform compatibility	5	5	5
Reliability	4	3	3
Vendor dependency	5	5	2
Offline/Low connectivity	3	3	2
User support	5	3	4
Total Score	**54**	**49**	**41**

4.2 Recommended Solution

Option 1, **Custom developed dedicated campus parking application: PukParking**, is the recommended solution with a score of **54/60**.

Key reasons for this recommendation:

1. **Alignment with user requirements:** A custom built solution will be designed specifically around the need identified through the users, instead of being constrained by the fixed-features of a third-party platform. This will allow for the solution to be tailored precisely to what the NWU student/staff/visitors require.

2. **NWU policy compliance:** A solution built in-house can fully align the access control, security measures with the internal IT governance and POPIA requirements of the NWU from the beginning rather than relying on the compliance of a third-party vendor in which the NWU has limited control over.

3. **Maintenance:** Maintenance of a custom-built solution can be scheduled and managed internally by the IT staff of the NWU, giving full control over system stability.

4. **Scalability**: The system can be designed from the get go to allow new parking zones, campuses or features to be added without needing a full redesign of the system.

5. **Platform compatibility**: A web-based system is accessible to any device with a browser regardless of the operating system it uses. It also maximizes accessibility because it does not require users to download or install a dedicated app.

6. **Vendor dependence**: Building the system in-house (or as a custom academic project) removes reliance on third-party vendors for core functionality, licensing, or pricing changes, reducing long-term risk of cost increases or service discontinuation that would affect a purchased or subscription-based alternative.

5 Implementation Approach

The implementation of the PukParking system follows the Design Science Research Methodology (DSRM) framework of Daase et al. (2022), which establishes the overarching research structure, combined with Agile development practices informed by the Agile Design Science Research Methodology (ADSRM) of Conboy et al. (2015:168).

The DSRM framework breaks the project down into 6 sequential activities, these steps being ; Problem Identification and Motivation, Definition of Objectives, Design and Development, Demonstration, Evaluation and Communication. During the development of this project two-week agile sprints are used to get regular feedback from the stakeholders and to do continuous testing.

This hybrid approach is valid because:

- **Operationalises the design cycle:** Agile sprints motivates build-and-test iterations (Hevner, 2007:89). This keeps the artifact aligned with theoretical design principles (Gregor & Jones, 2007:315).
- **Enables formative evaluation:** Short cycles detect design flaws early through ongoing testing (Venable et al., 2016:79). Frequent stakeholder feedback directly guides each revision (Sein et al., 2011:40).
- **Adapts to academic constraints:** Timeboxed sprints allow testing to pause during NWU exam periods. Tasks like documentation and refactoring can fill these blackout windows.

5.1 Project Initiation (DSR Activity 1–2: Problem Identification and Objectives Definition)

- **Define Project Scope:** PukParking is a login-free mobile first Progressive Web App (PWA) designed to deliver parking and pedestrian navigation recommendations across the NWU Potchefstroom campus.
- **Appoint Project Team:** Developer/researcher (Marco Engels), client representative (Me Selouna Visser, Project Sponsor), and NWU IT liaison (for infrastructure and card-system migration updates).
- **Establish Communication Plan:** Being governed by Agile and DSRM frame work there will be bi-weekly check-ups with the project sponsors and stakeholders ensuring direct alignment with the NWU broader digital transformation strategy whilst going through final user acceptance testing.
- **Conduct Literature Review:** This examines sensor less parking heuristic, GIS-based pathfinding and temporal decay modelling to enable navigation on campus through a PWA framework. It enforces privacy-by-design to ensure data privacy under the Protection of Personal Information Act (POPIA Act 4 of 2013).

5.2 Project Planning (DSR Activity 3a: Design)

- **Define System Architecture:** The system is designed in a 3-tier architecture mobile-first React, TypeScript and Tailwind CSS PWA with Service Workers for offline caching. Spatial layers are driven by Mapbox GL JS, MapBox API and device geolocation to handle the visual map on the user's device. The backend runs on a Node.js and Express.js REST API that sends parking data to a MongoDB database via Mongoose.
- **Geospatial & Campus Data Audit:** All NWU Potchefstroom campus parking areas, access boom gates, physical bay capacities, operating hours, and parking disc zone classifications (e.g., Yellow/Student, Red/Staff, Visitor, Accessible) will be catalogued and surveyed.
- **Algorithm & Service Configuration:** Create Parking recommendation logic and the crowd-sourced availability mode including a cooldown period of about 10-15 min. Set up the cloud service and the API keys to MongoDB and Mapbox.
- **Define Sprint Backlog:** Split app features into smaller bite sizes so users can customise where parking is shown to them for example first year students will select parking spots where their disks allow them to park.
- **Plan Sprint Schedule:** a Single sprint will last about 2 weeks and there will be about 6 sprints for the duration of the project
- **Define Acceptance Criteria:** Map Navigation, Current location, Direct map selection, Building name search (By name, abbreviation), view Building information, Nearest Valid Parking recommendation, Parking type and restriction filtering, Parking details inspection, Walking route generation, Distance and walking time ETA, Alternative route recalculation, Instant Parking Full report and Dynamic re-routing

5.3 Project Execution

Development will be completed incrementally. Testing will occur within each sprint instead of being postponed until the entire application is complete. This follows systems-development guidance that individual modules should be tested before integration and full system testing.

Proposed Execution Schedule

Sprint	Phase	Main Activity	Physical Deliverables
Sprint 0: 17-21 Aug	Foundation	Configure React + TypeScript + Vite, Express API, MongoDB, Git repository and environment variables	Running front-end, API and database

Sprint 1: 24 Aug-4 Sep	Campus Data & Admin	Create Buildings, Parking Areas and Administrator collections; implement JWT login and CRUD	Working administrator portal and campus database
Sprint 2: 7-18 Sep	Mapping & Search	Implement Mapbox GL JS, building/parking markers, geolocation and building name/abbreviation search	Interactive campus map and destination search
Sprint 3: 21 Sep-2 Oct	Recommendation & Navigation	Implement parking restriction filtering, Haversine distance calculation, ranking and Mapbox Directions API	Valid parking recommendation, walking route, distance and estimated walking time
Sprint 4: 5-9 Oct	Crowd Re-routing & PWA	Implement Parking Full reports, time-decay priority, alternative parking, service workers and offline caching	Crowd-assisted re-routing and installable PWA
Sprint 5: 12-16 Oct	Testing & Deployment	Integration, security, browser/mobile, performance and UAT testing; fix critical defects	Approved release and deployed PukParking MVP

The application will use the specified React/TypeScript front end, Express REST API, MongoDB and Mapbox APIs, with crowd-assisted parking reassignment rather than sensors.

5.4 Project Closure

- **Deliver the Final System:** Deploy the production PWA client and Express API to cloud hosting; configure custom domain/subdomain routing.
- **Launch Campus Access Points:** Generate and place promotional QR-code signage at campus entry gates/booms allowing instant mobile browser launching without app store downloads.
- **Provide User Documentation:** Deliver System Administration Guide, API Endpoint Documentation, and operational troubleshooting manuals to NWU Protection Services.
- **Conduct Administrator Training:** Deliver a focused training session (1–2 hours) for campus protection and facility staff on managing lot closures and reviewing parking traffic patterns.

- **Transfer Knowledge:** Provide the client with full access to the source code repository, deployment scripts, database schemas, and configuration keys.
- **Communicate Results:** Publish the DSR artefact and evaluation findings as part of the academic project deliverables, contributing design knowledge to the Information Systems research community regarding lightweight, crowdsourced campus navigation (Daase et al., 2022).

5.5 Project Management

Project management will operate throughout initiation, planning, execution and closure, while applying the following proccesses.

Process	Implementation
Time Management	Six execution stages will be tracked against the 16 October 2026 deadline. Sprint progress will be reviewed weekly.
Cost Management	MVP target is R0/near-zero. React, Node.js, Express and MongoDB are open source. Mapbox provides approximately 50 000 web map loads/month within the stated free tier; paid usage requires approval.
Quality Management	Each feature must meet its acceptance criteria and pass unit, integration and system testing before completion.
Change Management	Scope changes will be logged with reason, priority, effort and schedule impact before approval and addition to the backlog.
Risk Management	Key risks include inaccurate parking data, unreliable crowd reports, Mapbox/connectivity dependency, development delays and low adoption. Mitigation actions will be reviewed every sprint.
Issue Management	Actual problems will be recorded with priority, owner, target resolution date and status. High-priority issues will be resolved before release.
Procurement Management	No sensor hardware is required for the MVP. Mapbox is the main external technology supplier; any hosting or paid API services require approval.
Communications Management	Weekly team reviews and end-of-sprint demonstrations will report progress, risks, issues and decisions.
Acceptance Management	Features will be accepted during sprint reviews. Final acceptance requires successful UAT of search, mapping, parking recommendation, navigation, Parking Full re-routing, administrator functions and PWA operation.

This management approach supports the systems-project principle of delivering an acceptable system within the required time, cost and quality constraints.

6 Appendix

6.1 Supporting Documentation

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